

EM for censored poisson data

observed data

$$\text{Form 1: } y_1, \dots, y_m, y_{m+1} < 2, \dots, y_n < 2$$

$$\text{Form 2: } y_1, \dots, y_m, x_{m+1} = 1, \dots, x_n = 1$$

$$\text{where } x_i = I(y_i < 2)$$

Complete likelihood:

$$\begin{aligned} & f(y_1, \dots, y_m, y_{m+1}, \dots, y_n, x_{m+1}=1, \dots, x_n=1) \\ &= \prod_{i=1}^m f^{\text{Pois}}(y_i | \lambda) \cdot \prod_{i=m+1}^n f^{\text{Pois}}(y_i | \lambda) \cdot \prod_{i=m+1}^n I(y_i < 2, x_i=1) \end{aligned}$$

Observed likelihood:

$$\begin{aligned} & f(y_1, \dots, y_m, x_{m+1}=1, \dots, x_n=1) \\ &= \prod_{i=1}^m f^{\text{Pois}}(y_i | \lambda) \cdot \prod_{i=m+1}^n P(y_i < 2 | \lambda) \end{aligned}$$

$$\text{where } P(y_i < 2 | \lambda) = \sum_{y_i=0}^1 f^{\text{Pois}}(y_i | \lambda)$$

Let $Y = (y_1, \dots, y_m, \underbrace{y_{m+1}, \dots, y_n}_{\leftarrow \text{missing}})$

$$X = (x_{m+1}, \dots, x_n)$$

EM algorithm

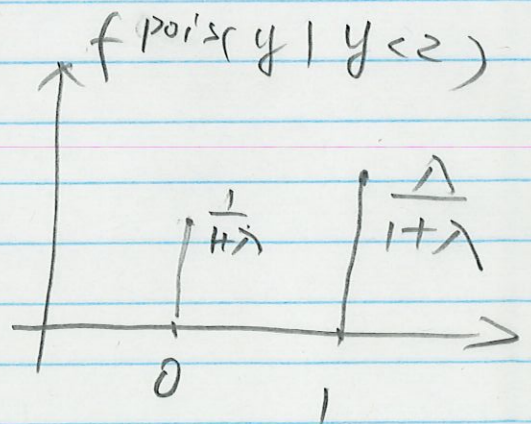
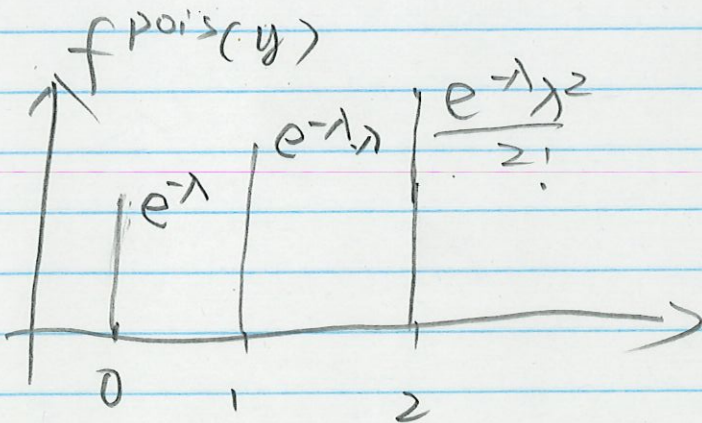
Given a previous estimate $\lambda^{(t)}$:

E-step: Compute $E_{Y|\lambda^{(t)}, X}(\log(f(Y, X|\lambda)))$

$$\begin{aligned} \ell^c(\lambda | Y, X) &= \sum_{i=1}^m \log\left(\frac{e^{-\lambda} \lambda^{y_i}}{y_i!}\right) + \sum_{i=m+1}^n \log\left(\frac{e^{-\lambda} \lambda^{y_i}}{y_i!}\right) + 0 \\ &= -(m+n)\lambda + \lambda \sum_{i=1}^m y_i + \lambda \sum_{i=m+1}^n y_i - \sum_{i=1}^n \log(y_i!) \end{aligned}$$

$$f(y_i | x_i=1, \lambda^{(t)}) = \begin{cases} \frac{e^{-\lambda^{(t)}}}{1 + \lambda^{(t)}} & , y_i = 0 \\ \frac{\lambda^{(t)}}{1 + \lambda^{(t)}} & , y_i = 1 \end{cases}$$

$\Downarrow$
 $y_i < 2$



$$E_{\lambda^{(t)}}(y_i | x_i=1) = \frac{\lambda^{(t)}}{1 + \lambda^{(t)}} \equiv y_i^{(t)}$$

$$Q(\lambda | \lambda^{(t)})$$

$$= E_{\lambda^{(m)}} \ell^c(\lambda | Y, X)$$

$$= -(m+n)\lambda + \lambda \sum_{i=1}^m y_i + \lambda \sum_{i=m+1}^n E_{\lambda^{(t)}}(y_i | x_i) + C^{(t)}$$

$$= -(m+n)\lambda + \lambda \sum_{i=1}^m y_i + \lambda \sum_{i=m+1}^n y_i^{(t)} + C^{(t)}$$

↑
unrelated
to λ

M-Step:

$$\lambda^{(t+1)} = \underset{\lambda}{\operatorname{argmax}} Q(\lambda | \lambda^{(t)})$$

$$= \frac{\sum_{i=1}^m y_i + \sum_{i=m+1}^n y_i^{(t)}}{m+n}$$

$$= \frac{m \bar{y}_{1:m} + (n-m) \frac{\lambda^{(t)}}{1+\lambda^{(t)}}}{m+n}$$